

Digit puzzles in the classroom

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Digit puzzles

Given a collection of digits/numbers and allowable operations, make a target number.

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Given a collection of digits/numbers and allowable operations, make a target number.

Unless otherwise specified, brackets and reordering are allowed.

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24

Flip over 4 cards

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Flip over 4 cards

- $A = 1$

24

Flip over 4 cards

- $A = 1$
- $J = 11$

24

Flip over 4 cards

- $A = 1$
- $J = 11$
- $Q = 12$

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- $A = 1$
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Use standard operations $(+, -, \times, \div)$ to make 24

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Use standard operations $(+, -, \times, \div)$ to make 24

First person takes set of cards as points

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Flip over 4 cards

- $A = 1$
- $J = 11$
- $Q = 12$
- $K = 13$

Use standard operations ($+$, $-$, $\times$, $\div$) to make 24

First person takes set of cards as points

Mistake = lose a set

Examples

2, 4, 8, 10

Examples

2, 4, 8, 10

$$2 + 4 + 8 + 10$$

Examples

2, 4, 8, 10

$$2 + 4 + 8 + 10$$

1, 5, 10, 13

Examples

2, 4, 8, 10

$$2 + 4 + 8 + 10$$

1, 5, 10, 13

$$10 \div 5 \times (13 - 1)$$

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2, 4, 8, 10

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$$10 \div 5 \times (13 - 1)$$

1, 3, 4, 6

Examples

2, 4, 8, 10

$$2 + 4 + 8 + 10$$

1, 5, 10, 13

$$10 \div 5 \times (13 - 1)$$

1, 3, 4, 6

$$6 / (1 - 3/4)$$

Examples

2, 4, 8, 10

$$2 + 4 + 8 + 10$$

1, 5, 10, 13

$$10 \div 5 \times (13 - 1)$$

1, 3, 4, 6

$$6 / (1 - 3/4)$$

3, 4, 9, 10

Examples

2, 4, 8, 10

$$2 + 4 + 8 + 10$$

1, 5, 10, 13

$$10 \div 5 \times (13 - 1)$$

1, 3, 4, 6

$$6 / (1 - 3/4)$$

3, 4, 9, 10

Impossible

$\approx 75\%$ chance of solution

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Three options when someone calls “impossible”:

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- Everyone can agree to skip a set of cards.

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Three options when someone calls “impossible”:

- Everyone can agree to skip a set of cards.
- A timer starts: during this time, if someone solves it, the solver wins the set and the caller loses a set. If no one solves by the end of the time, the caller wins the set.

$\approx 75\%$ chance of solution

Three options when someone calls “impossible”:

- Everyone can agree to skip a set of cards.
- A timer starts: during this time, if someone solves it, the solver wins the set and the caller loses a set. If no one solves by the end of the time, the caller wins the set.
- Use an online solver to check if a solution exists. If one does, the caller loses a set.

Try it out!

- 1, 9, 12, 12
- 2, 8, 9, 12
- 3, 9, 11, 12
- 4, 7, 11, 11
- 5, 9, 12, 13
- 6, 8, 12, 13
- 7, 8, 8, 13
- 8, 10, 12, 12
- 9, 10, 12, 13
- 10, 11, 13, 13

Try it out!

- 1, 9, 12, 12: $(12 - 9 - 1) \times 12$
- 2, 8, 9, 12: $(9 - 8 + 1) \times 12$
- 3, 9, 11, 12: $(9 + 11 - 12) \times 3$
- 4, 7, 11, 11: $(7 - 11 \div 11) \times 4$
- 5, 9, 12, 13: $(12 - 9) \times (13 - 5)$
- 6, 8, 12, 13: impossible
- 7, 8, 8, 13: $(8 + 13) \div 7 \times 8$
- 8, 10, 12, 12: $12 \times 10 - 12 \times 8$
- 9, 10, 12, 13: $12 + 13 - 10 + 9$
- 10, 11, 13, 13: impossible

Variations

Exponent 24

In addition to usual 24 rules, must use 1 of:

$$\square^2, \square^3, \square^{1/2}, \square^{1/3}$$

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- Introduces powers and radicals

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Cons:

- No interesting solutions (yet)

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$$\square^2, \square^3, \square^{1/2}, \square^{1/3}$$

Pros:

- Introduces powers and radicals
- Increases probability of solution to 97%

Cons:

- No interesting solutions (yet)
- Can be nice to check for impossibility

Variations

163

Flip over 6 cards. Using standard operations, make 163.

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Pros:

Variations

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- Much higher complexity (including some interesting solutions)

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Variations

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- Much higher complexity (including some interesting solutions)
- Increases probability of solution to 99%

Cons:

- Much higher complexity

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Using 3 given digits, make a given target number. Brackets are allowed, and you can use (finitely many of) the following operations:

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- Standard operations: $+$, $-$, $\times$, $\div$
- Negation: $-\square$
- Exponentiation of two numbers: $\square^{\square}$

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- Factorial: $\square!$

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- Factorial: $\square!$
- Digit decimal and digit repeating decimal: $.d$ and $.\overline{d}$ (with an original digit; you cannot do $.4! = .24$)

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Using 3 given digits, make a given target number. Brackets are allowed, and you can use (finitely many of) the following operations:

- Standard operations: $+$, $-$, $\times$, $\div$
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- Square root and n th root: $\sqrt{\square}$ and $\sqrt[n]{\square}$
- Factorial: $\square!$
- Digit decimal and digit repeating decimal: $.d$ and $.\overline{d}$ (with an original digit; you cannot do $.4! = .24$)
- Percent: $\square\%$

Examples

Some puzzles are easy:

1, 3, 4 to make 5

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1, 3, 4 to make 5

$$1^3 + 4$$

Examples

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1, 3, 4 to make 5

$$1^3 + 4$$

Some take a bit of thought:

1, 1, 4 to make 11

Examples

Some puzzles are easy:

1, 3, 4 to make 5

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Some take a bit of thought:

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$$\sqrt{1 + (1 + 4)!}$$

Examples

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1, 3, 4 to make 5

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Some take a bit of thought:

1, 1, 4 to make 11

$$\sqrt{1 + (1 + 4)!}$$

Some are incredibly hard:

2, 4, 4 to make 17

Examples

Some puzzles are easy:

1, 3, 4 to make 5

$$1^3 + 4$$

Some take a bit of thought:

1, 1, 4 to make 11

$$\sqrt{1 + (1 + 4)!}$$

Some are incredibly hard:

2, 4, 4 to make 17

$$2 + {}^{-\sqrt{4}}\sqrt{{}^{\sqrt{4}}.4\%}$$

Original game

Standard game:

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Standard game:

- Get three random digits 1 to 6 by rolling dice

Original game

Standard game:

- Get three random digits 1 to 6 by rolling dice
- Try to get a target number from 0 to 20 (or all of the numbers)

Try it out!

Make the numbers 0 to 20 using the digits 1, 2, 3. If you want a challenge, try to do it with the digits in order!

- | | | |
|-----|------|------|
| • 0 | • 7 | • 14 |
| • 1 | • 8 | • 15 |
| • 2 | • 9 | • 16 |
| • 3 | • 10 | • 17 |
| • 4 | • 11 | • 18 |
| • 5 | • 12 | • 19 |
| • 6 | • 13 | • 20 |

Try it out!

Make the numbers 0 to 20 using the digits 1, 2, 3. If you want a challenge, try to do it with the digits in order!

$$\bullet 0 = 1 + 2 - 3$$

$$\bullet 1 = 1^{2+3}$$

$$\bullet 2 = 1 - 2 + 3$$

$$\bullet 3 = 1^2 \times 3$$

$$\bullet 4 = 1^2 + 3$$

$$\bullet 5 = 1 \times 2 + 3$$

$$\bullet 6 = 1 + 2 + 3$$

$$\bullet 7 = 1 + 2 \times 3$$

$$\bullet 8 = 1 \times 2^3$$

$$\bullet 9 = 1 + 2^3$$

$$\bullet 10 = (1 + 2) \div .3$$

$$\bullet 11 = -1 + 2 \times 3!$$

$$\bullet 12 = 1 \times 2 \times 3!$$

$$\bullet 13 = 1 + 2 \times 3!$$

$$\bullet 14 = 2 \times (1 + 3!)$$

$$\bullet 15 = 1 \div .2 \times 3$$

$$\bullet 16 = 2^{1+3}$$

$$\bullet 17 = 2 \div .1 - 3$$

$$\bullet 18 = (1 + 2) \times 3!$$

$$\bullet 19 = \sqrt{3!! \div 2 + 1}$$

$$\bullet 20 = (1 + 3) \div .2$$

Try it out!

Challenge:

• $0 = 1 + 2 - 3$

• $1 = 1^{2+3}$

• $2 = 1 - 2 + 3$

• $3 = 1^2 \times 3$

• $4 = 1^2 + 3$

• $5 = 1 \times 2 + 3$

• $6 = 1 + 2 + 3$

• $7 = 1 + 2 \times 3$

• $8 = 1 \times 2^3$

• $9 = 1 + 2^3$

• $10 = (1 + 2) \div .3$

• $11 = -1 + 2 \times 3!$

• $12 = 1 \times 2 \times 3!$

• $13 = 1 + 2 \times 3!$

• $14 = (\sqrt{.1} + 2) \times 3!$

• $15 = 1 \div .2 \times 3$

• $16 = .1 \times .\bar{2} \times 3!!$

• $17 = (1 + 2\%) \div 3!\%$

• $18 = (1 + 2) \times 3!$

• $19 = ?$

• $20 = 1 \div (2 + 3)\%$

Variations

Use a random number generator to get digits from 1 to 9 or cards to get numbers from 1 to 13 instead of 1 to 6

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Try to reach a target number from 0 to 100 (or all of the numbers)

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Use more than 3 numbers (e.g. 4 dice)

Variations

Use a random number generator to get digits from 1 to 9 or cards to get numbers from 1 to 13 instead of 1 to 6

Try to reach a target number from 0 to 100 (or all of the numbers)

Restrict to using the digits in order

Use more than 3 numbers (e.g. 4 dice)

Experiment with other operations or restrict operations

Variations

3 dice target

Get as close as possible to a target number from 0 to 100.

Variations

3 dice target

Get as close as possible to a target number from 0 to 100.
First person to the closest value within a time limit wins a point.

Variations

3 dice target

Get as close as possible to a target number from 0 to 100.
First person to the closest value within a time limit wins a point.

Variant:

- 3 points for closest integer
- 2 points for closest rational
- 1 point for closest real

Variations

4 dice (Complexity variant)

Using four given digits, make the numbers 0 to 20 using
 $+$, $-$, $\times$, $\div$, $\square^{\square}$, $\sqrt{\square}$, $\square!$.

For each target number:

Variations

4 dice (Complexity variant)

Using four given digits, make the numbers 0 to 20 using
 $+$, $-$, $\times$, $\div$, $\square^{\square}$, $\sqrt{\square}$, $\square!$.

For each target number:

- 1 point for a correct solution.

Variations

4 dice (Complexity variant)

Using four given digits, make the numbers 0 to 20 using $+$, $-$, $\times$, $\div$, $\square^\square$, $\sqrt{\square}$, $\square!$.

For each target number:

- 1 point for a correct solution.
- For each of $\div$, $\square^\square$, $\sqrt{\square}$, $\square!$, earn 1 point for the first usage, $\frac{1}{2}$ points for the 2nd, $\frac{1}{4}$ for the 3rd, and so on.

Variations

4 dice (Complexity variant)

Using four given digits, make the numbers 0 to 20 using $+$, $-$, $\times$, $\div$, $\square^\square$, $\sqrt{\square}$, $\square!$.

For each target number:

- 1 point for a correct solution.
- For each of $\div$, $\square^\square$, $\sqrt{\square}$, $\square!$, earn 1 point for the first usage, $\frac{1}{2}$ points for the 2nd, $\frac{1}{4}$ for the 3rd, and so on.

In a solution is unchanged by removing one more more unary operators, those operators don't count for points.

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Redigital puzzle

Using only copies of a single digit, together with $+$, $-$, $\times$, $\div$, $\square^\square$, $\sqrt{\square}$, $\square!$, make a target number.

Redigital puzzle

Using only copies of a single digit, together with $+$, $-$, $\times$, $\div$, $\square^\square$, $\sqrt{\square}$, $\square!$, make a target number. Concatenating the original digit is allowed!

Goal is to minimize the number of copies.

Examples

Make 2026 with 7s

Examples

Make 2026 with 7s

Easy:



Examples

Make 2026 with 7s

Easy:

2027 copies

$$\frac{7 + 7 + \cdots + 7}{7}$$

Examples

Make 2026 with 7s

Easy:

2027 copies

$$\frac{7 + 7 + \cdots + 7}{7}$$

Moderate:

Examples

Make 2026 with 7s

Easy:

2027 copies

$$\frac{7 + 7 + \cdots + 7}{7}$$

Moderate:

6 copies so far

$$777 + 777 = 1554$$

Examples

Make 2026 with 7s

Easy:

2027 copies

$$\frac{7 + 7 + \cdots + 7}{7}$$

Moderate:

9 copies so far

$$777 + 777 + 77 \times 7 = 2093$$

Examples

Make 2026 with 7s

Easy:

2027 copies

$$\frac{7 + 7 + \cdots + 7}{7}$$

Moderate:

11 copies so far

$$777 + 777 + 77 \times 7 - 77 = 2016$$

Examples

Make 2026 with 7s

Easy:

2027 copies

$$\frac{7 + 7 + \cdots + 7}{7}$$

Moderate:

16 copies

$$777 + 777 + 77 \times 7 - 77 + 7 + \frac{7 + 7 + 7}{7}$$

Examples

Make 2026 with 7s
Challenging:

10 copies

$$\frac{7! + 7! - 7 - 7}{7} + 77 \times 7 + 7 \times 7$$

Examples

Make 2026 with 7s
Challenging:

10 copies

$$\frac{7! + 7! - 7 - 7}{7} + 77 \times 7 + 7 \times 7$$

Insane:

Examples

Make 2026 with 7s
Challenging:

10 copies

$$\frac{7! + 7! - 7 - 7}{7} + 77 \times 7 + 7 \times 7$$

Insane:

8 copies

$$\left(\frac{\sqrt{\sqrt{7!}}}{\sqrt{\sqrt{7}} + \sqrt{\sqrt{7}}} \right)^{7+7/7} + \frac{7}{7}$$

Try it out!

Make 26 with as few copies as possible of each digit.

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9

Try it out!

Make 26 with as few copies as possible of each digit.

- 1 - 6 copies: $(1 + 1 + 1 + 1)! + 1 + 1$
- 2 - 3 copies: $(2 + 2)! + 2$
- 3 - 4 copies: $3!/(3! \times 3!) + 3!$
- 4 - 2 copies: $4! + \sqrt{4}$
- 5 - 4 copies: $5 \times 5 + 5/5$
- 6 - 4 copies: $6!/(6 \times 6) + 6$
- 7 - 6 copies: $(77 + 7)/7 + 7 + 7$
- 8 - 4 copies: $\sqrt{8 + 8!} + \sqrt{\sqrt{8 + 8}}$
- 9 - 4 copies: $9 \times \sqrt{9} - 9/9$

Conclusion

Simple to learn, but challenging to master

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Highly differentiable and engaging

Conclusion

Simple to learn, but challenging to master

Highly differentiable and engaging

Mathematics is not memorizing algorithms, but about playing with ideas

Thank you for listening!

Questions?

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